

W241 Plank Study Experimental Report

Crossover and benchmark analysis

W241 Plank Study

2026-04-12

Contents

0.1	Report Structure	2
0.2	Study Overview	2
0.3	Participant Flow And Analysis Populations	2
0.4	Reading the flow correctly	2
0.5	Build participant flow dataset	2
0.6	Participant flow summaries	3
0.7	Build paired analytic dataset	3
0.8	Build long repeated-measures dataset	4
0.9	Write derived outputs	4
0.10	Primary Crossover Estimate	4
0.11	Analytic sample	4
0.12	Descriptive summaries	4
0.13	Compact metrics summary	4
0.14	Exploratory figures	5
0.15	Model 1: raw paired contrast	6
0.16	Model 2: fixed-effects crossover regression	6
0.17	Main-effect comparison figure	7
0.18	Period And Order Diagnostics	7
0.19	Period summaries	7
0.20	Period-by-order summaries	7
0.21	Session-period figures	8
0.22	Diagnostic models	9
0.23	Interpretation notes	10
0.24	First-Period Benchmark	10
0.25	Benchmark sample summary	10
0.26	Balance checks	10
0.27	Benchmark descriptives	11
0.28	Figure	11
0.29	Benchmark Model 1: raw contrast	11
0.30	Benchmark Model 2: precision-adjusted regression	11
0.31	Results At A Glance	12
0.32	Precision And Design-Efficiency Comparison	12
0.33	Precision calculations	13
0.34	Precision table	13
0.35	Precision comparison figure	13
0.36	Interpretation notes	13
0.37	Exploratory Mechanisms And Heterogeneity	14
0.38	Secondary outcomes: paired contrasts	14

0.39	Secondary outcomes: fixed-effects versions	14
0.40	HTE setup	14
0.41	HTE models	14
0.42	Design-Based Inference Check	14
0.43	Observed statistic	15
0.44	Randomization distribution	15
0.45	Figure	15

0.1 Report Structure

This report follows the order in the experiment analysis plan:

1. participant flow, attrition, and exclusions
2. primary crossover effect
3. period and order diagnostics
4. first-period benchmark
5. realized precision comparison
6. secondary outcomes and HTE
7. randomization inference

0.2 Study Overview

The primary estimand is the **within-person effect of high-tempo versus low-tempo music on plank time-to-exhaustion**, measured in seconds. The main estimate uses the randomized crossover design, so each participant serves as their own control and stable participant-level differences are absorbed by construction.

The first-period benchmark is supplementary. It asks what the treatment contrast would have looked like if the study had been analyzed as a standard parallel-arm design using only session 1. That benchmark is included to clarify the design choice, not to replace the crossover estimate.

0.3 Participant Flow And Analysis Populations

The report begins with the full registered participant roster, then narrows to observed outcomes and protocol-valid analytic samples. This keeps attrition, missingness, and protocol exclusions visible before any inferential models are introduced.

0.4 Reading the flow correctly

This step intentionally starts from the **full initial distribution** and only then narrows to the analytic samples.

- First, we show the full registered roster and how many participants reach each observation stage.
- Next, we separate **attrition** (for example, not returning for session 2) from **compliance / protocol exclusions**.
- Only after those counts are visible do we define the **primary analytic sample** and the **first-period benchmark sample**.

This mirrors the report plan: the reader should see the whole study population first, then exactly why some participants are not used for a given estimand.

0.5 Build participant flow dataset

The participant roster is the base denominator for flow accounting. We left-join the experiment wide file onto the full participant list so registered participants with zero observed sessions still appear in the flow dataset.

0.6 Participant flow summaries

Table 1: Table A. Participant flow

population	n
Registered participants	58
Randomized order assigned	58
Session 1 observed	47
Session 2 observed	40
Completed both sessions	40
Primary analytic sample	38
First-period benchmark sample	46

Participant flow from registration to analytic samples

Full distribution first, then attrition and protocol-based exclusions

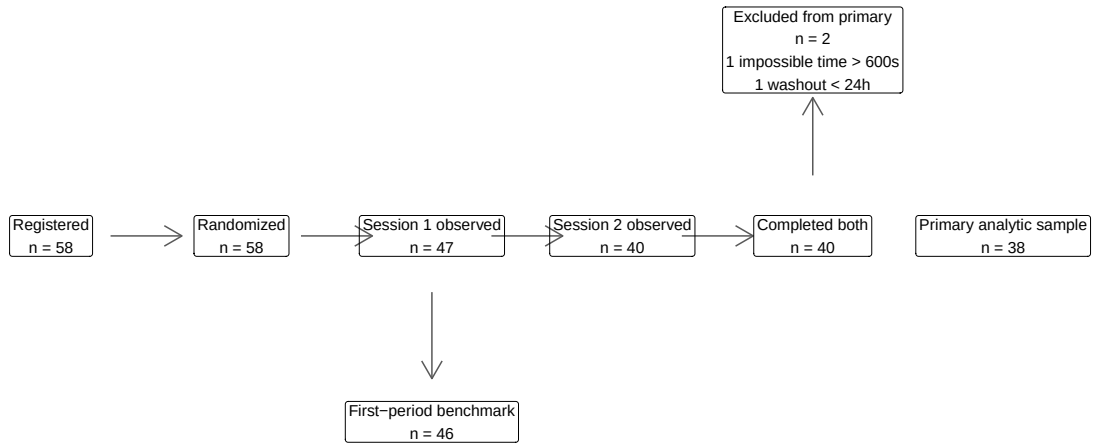


Table 2: Table B. Attrition by randomized order

assigned_order	started_session1	dropped_before_session2	retained_completed_both
high_tempo_first	23	4	19
low_tempo_first	24	3	21

Table 3: Table C. Compliance and exclusion reasons

issue	n
Impossible time > 600s	1
Washout violation < 24h	1
Missing one condition	18
Excluded from primary analysis	2

0.7 Build paired analytic dataset

The paired dataset keeps one row per primary analytic participant and standardizes condition-level secondary outcomes so later Rmds can work in `high_tempo` / `low_tempo` terms directly.

0.8 Build long repeated-measures dataset

The long dataset keeps one row per observed session and carries analysis flags so later model files can subset consistently without re-implementing the exclusion logic.

0.9 Write derived outputs

0.10 Primary Crossover Estimate

With the populations fixed, the next step is the primary within-subject treatment estimate. The emphasis here is the design-consistent sequence: paired descriptives first, then the paired mean difference, then the fixed-effects crossover model with a period adjustment.

0.11 Analytic sample

Table 4: Primary crossover sample

quantity	value
Paired analytic participants	38
Repeated-measures rows used in FE model	76
High-tempo first	19
Low-tempo first	19

0.12 Descriptive summaries

Table 5: Paired descriptive summaries

quantity	value
Mean high-tempo TTE	102.440
Mean low-tempo TTE	88.071
Mean paired difference (high - low)	14.369
SD paired difference	28.921
Median paired difference	8.708
IQR paired difference	25.199
Independent participants	38.000

0.13 Compact metrics summary

Table 6: Requested summary metrics

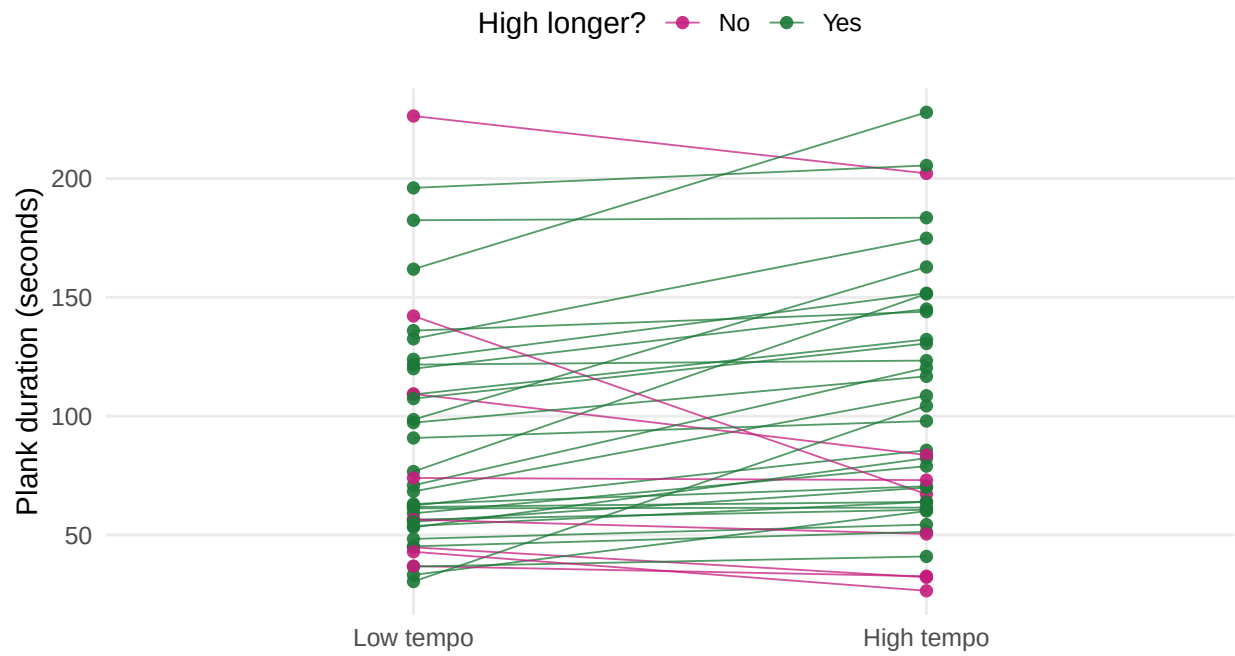
metric	value
Mean treatment effect (high - low, seconds)	14.37
Percentage improvement vs low tempo mean	16.3%
Standard deviation of the paired difference	28.92
Paired t-test p-value	0.0041
Completion rate (completed both / registered)	69.0%

0.14 Exploratory figures

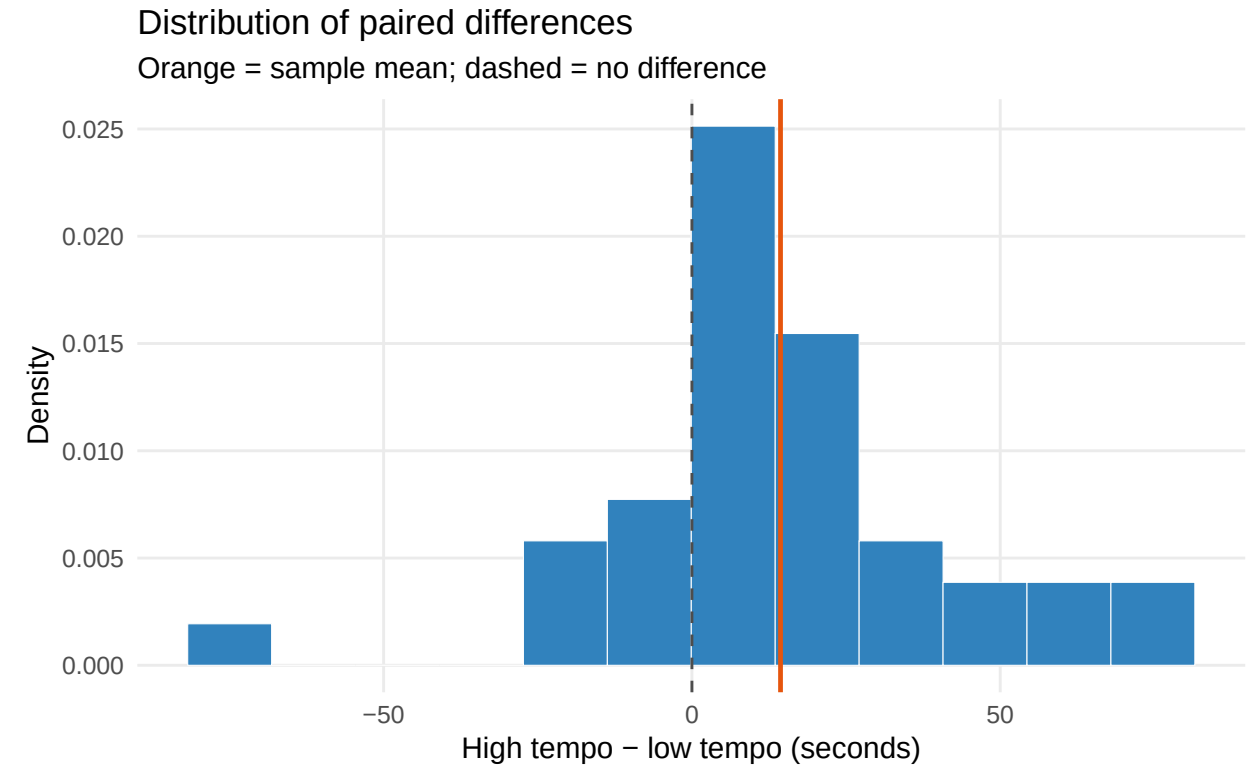
0.14.1 Spaghetti plot

Paired plank times by music condition

n = 38 primary-sample participants



0.14.2 Paired-difference distribution



0.15 Model 1: raw paired contrast

Table 7: Model 1. Raw paired contrast

quantity	value
Mean high-tempo TTE	102.4404
Mean low-tempo TTE	88.0709
Mean effect (high - low)	14.3694
95% CI lower	4.8633
95% CI upper	23.8756
Paired t statistic	3.0628
Paired p-value	0.0041
Standard error	4.6916
Paired effect size dz	0.4968

0.16 Model 2: fixed-effects crossover regression

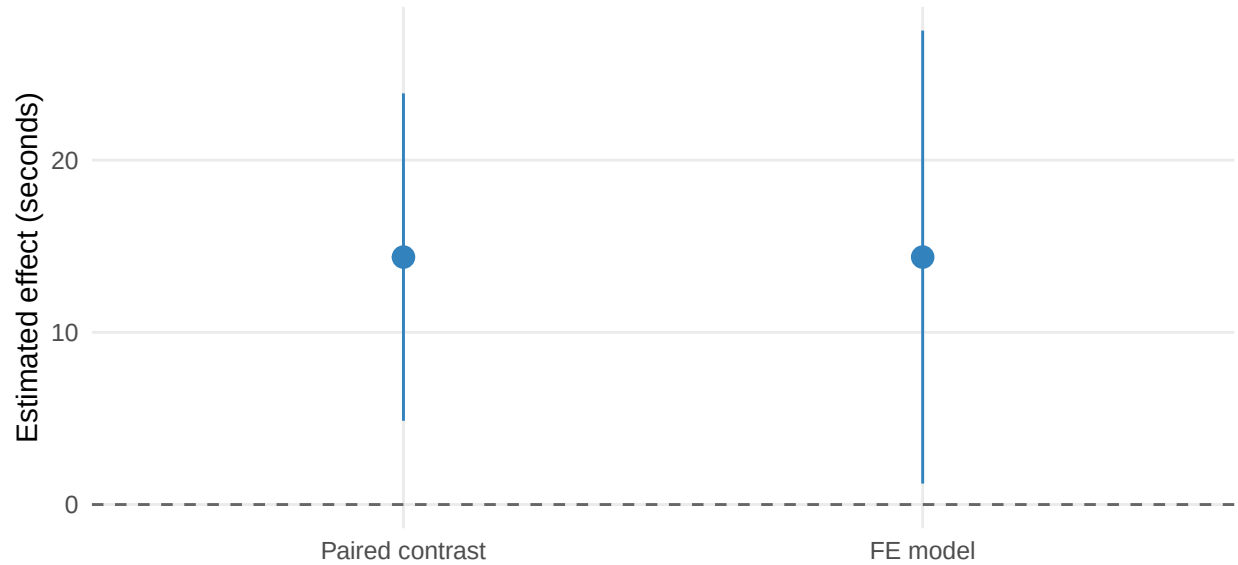
Table 8: Model 2. Fixed-effects crossover regression with clustered SEs

term	estimate	std_error	conf_low	conf_high	p_value
High-tempo indicator	14.3694	6.7101	1.2177	27.5212	0.0391
Period 2 indicator	3.8441	6.7101	-9.3076	16.9958	0.5703

0.17 Main-effect comparison figure

Primary effect estimates

Effect on plank time in seconds (high tempo – low tempo)



0.18 Period And Order Diagnostics

After the main estimate, the report checks whether broad period effects or order patterns appear large enough to complicate interpretation. These diagnostics are supportive rather than alternative estimators.

0.19 Period summaries

Table 9: Session 1 vs Session 2 summaries

period	n	mean_tte	sd_tte	median_tte	iqr_tte
1	38	93.334	46.730	84.646	68.303
2	38	97.178	55.207	70.692	75.268

0.20 Period-by-order summaries

Table 10: Period-by-order diagnostic summary

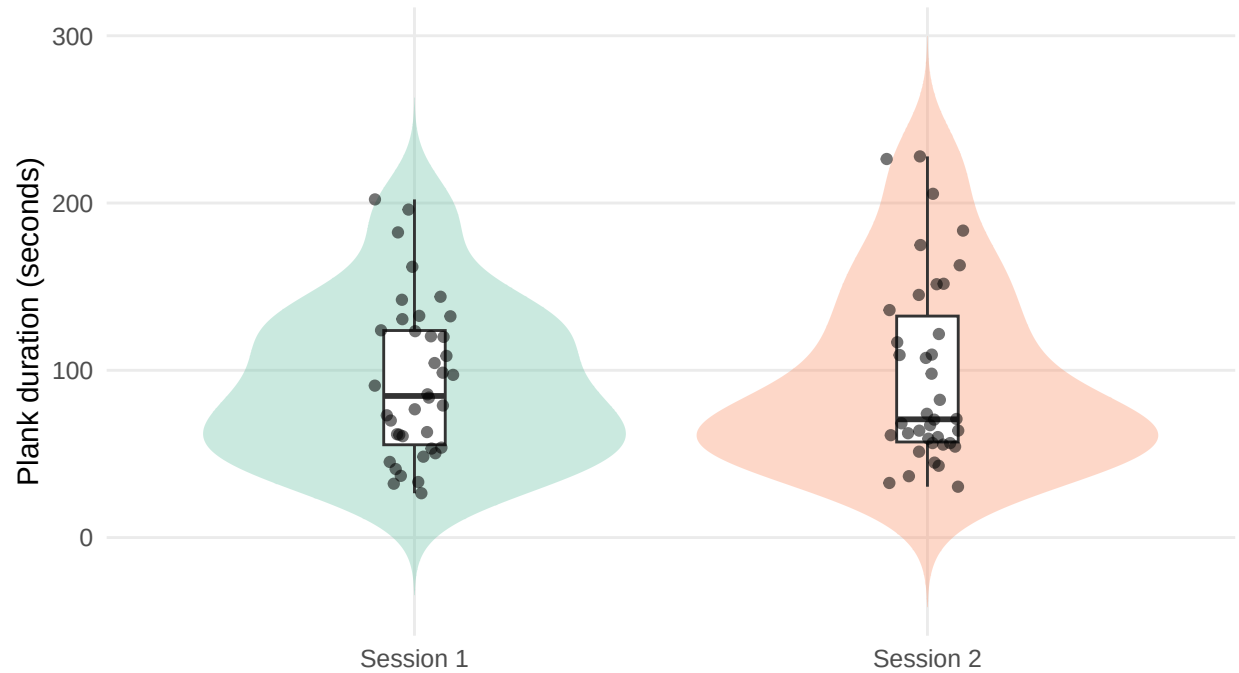
crossover_order	period	treatment	n	mean_tte	sd_tte	mean_rpe
high_tempo_first	1	high_tempo	19	91.001	44.336	6.895
high_tempo_first	2	low_tempo	19	80.476	46.395	6.263
low_tempo_first	1	low_tempo	19	95.666	50.116	7.105
low_tempo_first	2	high_tempo	19	113.880	59.359	7.632

0.21 Session-period figures

0.21.1 Session 1 vs Session 2 boxplot

Outcome distributions by session period

Primary analytic sample, n = 38



0.21.2 Order-arm trajectories across periods

Average plank time by order arm across periods

Checks for sequence or learning patterns



0.22 Diagnostic models

0.22.1 Broad period check

Table 11: Broad period comparison

quantity	value
Mean Session 2	97.1777
Mean Session 1	93.3336
Difference (Session 2 - Session 1)	3.8441
95% CI lower	-19.5456
95% CI upper	27.2339
p-value	0.7441

0.22.2 Sequence diagnostic regression

Table 12: Sequence diagnostic regression

term	estimate	std_error	p_value
period_factor2	-10.5253	5.8264	0.0750
crossover_orderlow_tempo_first	4.6650	15.4541	0.7636
period_factor2:crossover_orderlow_tempo_first	28.7389	9.4895	0.0034

0.23 Interpretation notes

0.24 First-Period Benchmark

Only after the primary crossover evidence and the period/order checks do we turn to the first-period benchmark. This keeps the benchmark in its intended role: a secondary comparison that illustrates what a parallel-arm analysis would have looked like before carryover could matter.

0.25 Benchmark sample summary

Table 13: First-period benchmark sample

quantity	value
Benchmark participants	46
Assigned high tempo first	23
Assigned low tempo first	23

0.26 Balance checks

Table 14: Compact randomization check artifact: counts, age, gender, and baseline activity

treatment_label	n	mean_age	share_male	share_active_at_least_weekly
High tempo first	23	31.17	73.9	34.8
Low tempo first	23	40.35	60.9	47.8

Table 15: Balance check: age by assigned first condition

treatment_label	n	mean_age
High tempo first	23	31.174
Low tempo first	23	40.348

Table 16: Balance check: gender counts

treatment_label	female	male
High tempo first	6	17
Low tempo first	9	14

Table 17: Balance check: baseline activity counts

treatment_label	1	2	3	4
High tempo first	9	6	1	7
Low tempo first	7	5	0	11

The first-period benchmark arms are balanced in **counts** but not perfectly identical on observables. The main age difference is visible here, which is worth acknowledging. It is less threatening for the primary estimate because the main analysis uses the crossover design and participant fixed effects, which absorb stable participant-level differences.

0.27 Benchmark descriptives

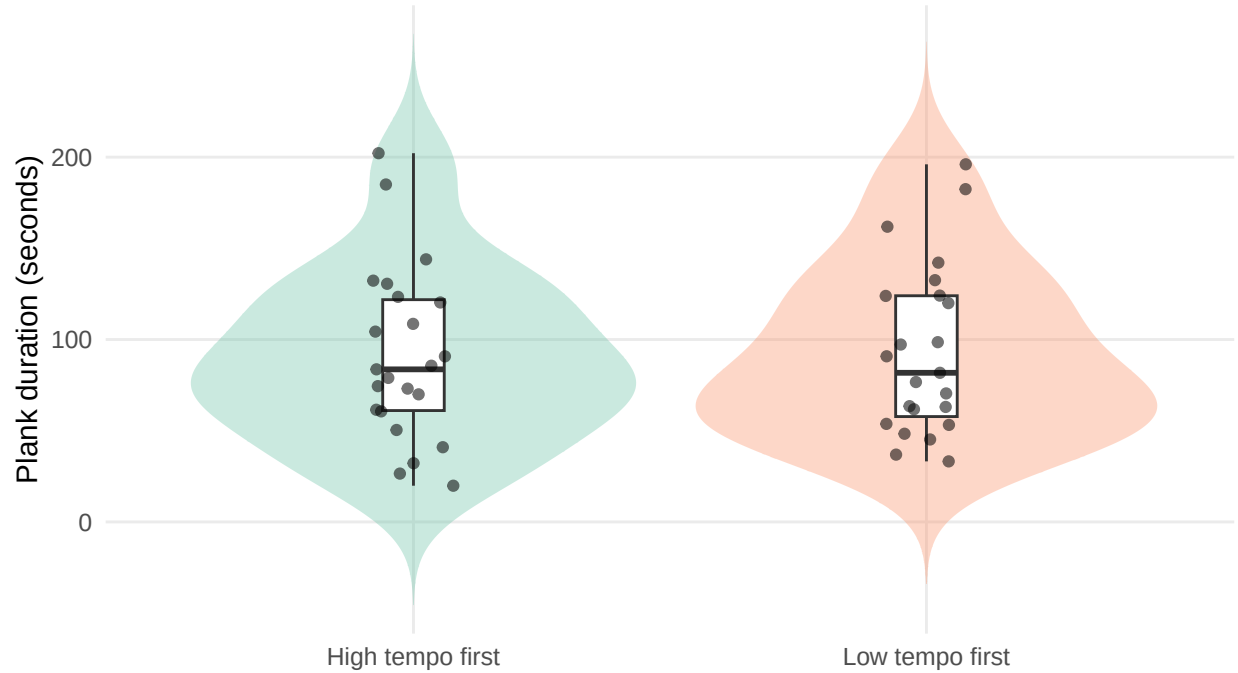
Table 18: Session 1 descriptives by assigned condition

treatment_label	n	mean_tte	sd_tte	median_tte	iqr_tte
High tempo first	23	91.265	47.460	83.655	60.795
Low tempo first	23	93.801	46.614	81.760	66.236

0.28 Figure

Session 1 benchmark by assigned first condition

n = 46 independent participants



0.29 Benchmark Model 1: raw contrast

Table 19: Benchmark Model 1. Raw period-1 contrast

quantity	value
Mean high-first session 1 TTE	91.2654
Mean low-first session 1 TTE	93.8013
Difference (high first - low first)	-2.5359
95% CI lower	-25.4194
95% CI upper	30.4912
t statistic	0.1828
p-value	0.8558

0.30 Benchmark Model 2: precision-adjusted regression

Table 20: Benchmark Model 2. Precision-adjusted regression

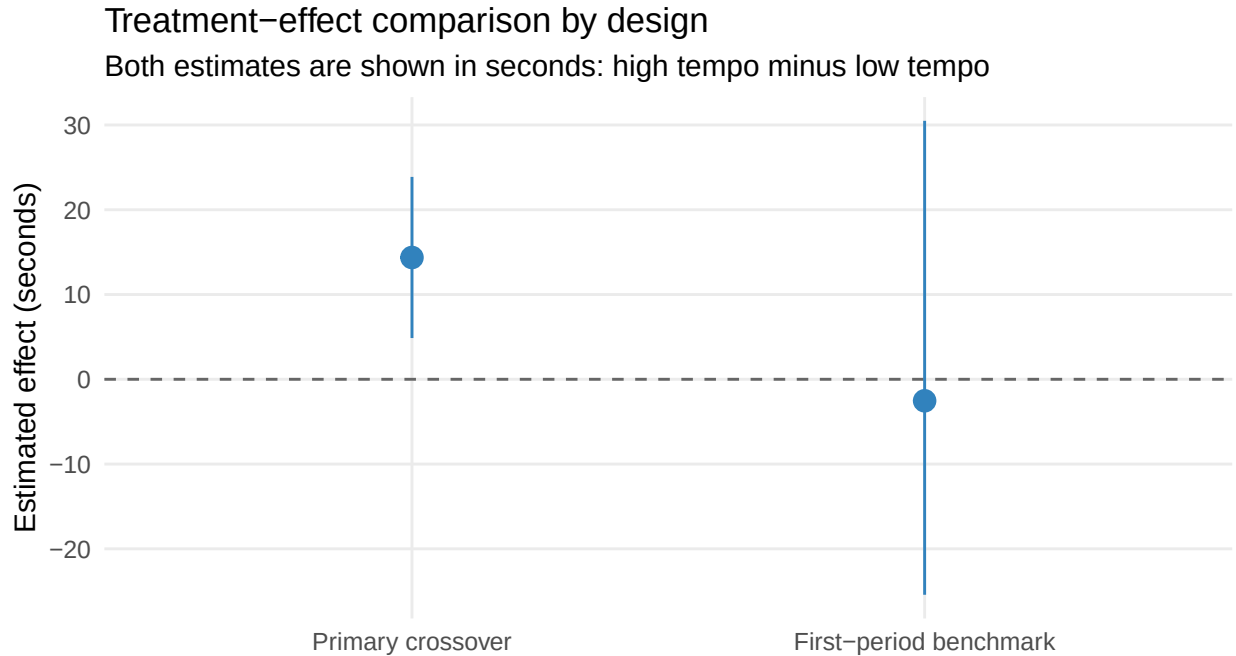
term	estimate	std_error	p_value
High-tempo first	-6.6796	15.0537	0.6595
Participant age	-0.1142	0.6001	0.8499

0.31 Results At A Glance

The key treatment-effect results are consolidated here so the reader can compare the primary crossover estimate, the supplementary benchmark, and the design-based inference in one place.

Table 21: Compact main-results summary

statistic	value
Paired analytic participants	38
Mean low-tempo TTE	88.07
Mean high-tempo TTE	102.44
Paired effect (high - low)	14.37
Paired 95% CI	[4.86, 23.88]
Paired p-value	0.0041
FE treatment estimate	14.37
FE p-value	0.0391
RI two-sided p-value	0.0031



0.32 Precision And Design-Efficiency Comparison

Once both the crossover and benchmark estimates are on the page, the precision comparison shows why the crossover design is more efficient for this question.

0.33 Precision calculations

0.34 Precision table

Table 22: Realized precision by design

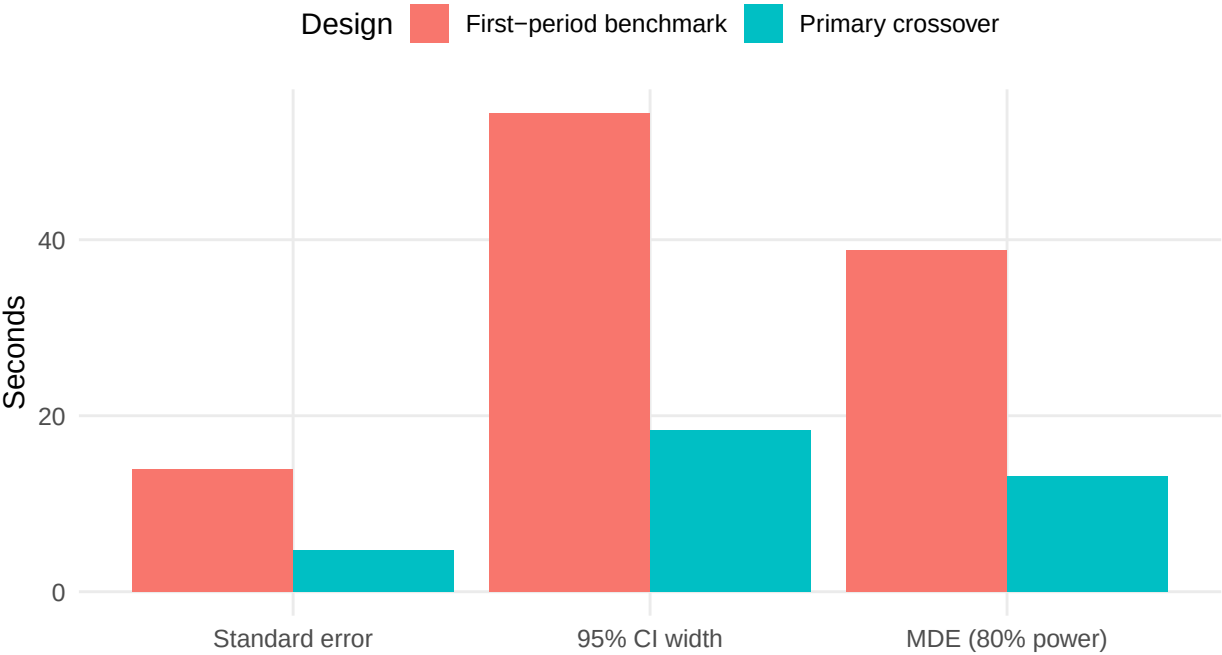
design	independent_n	observations	relevant_sd	standard_error	ci_width_95	mde_80_power
Primary crossover	38	76	28.9211	4.6916	18.3908	13.1440
First-period benchmark	46	46	47.0386	13.8709	54.3730	38.8606

Table 23: Parallel-design sample size needed to match crossover precision

quantity	value
Observed benchmark total N	46
Required benchmark total N to match crossover SE	404
Required benchmark N per arm to match crossover SE	202
Extra independent participants needed	358

0.35 Precision comparison figure

Realized precision comparison
Smaller values indicate tighter design precision



0.36 Interpretation notes

Table 24: Relative precision loss in the benchmark design

quantity	value
Parallel SE / crossover SE	2.9565
Parallel CI width / crossover CI width	2.9565

0.37 Exploratory Mechanisms And Heterogeneity

The next section keeps the mechanism-related outcomes and subgroup patterns clearly downstream of the primary estimate. These are exploratory checks rather than headline findings.

0.38 Secondary outcomes: paired contrasts

Table 25: Exploratory paired contrasts for secondary outcomes

outcome	mean_high	mean_low	mean_diff_high_minus_low	conf_low	conf_high	p_value
RPE	7.2632	6.6842	0.5789	-0.0161	1.1740	0.0562
Motivation	5.1842	4.7105	0.4737	0.0298	0.9176	0.0371
Energy	4.6053	4.3158	0.2895	-0.1748	0.7537	0.2144

0.39 Secondary outcomes: fixed-effects versions

Table 26: Exploratory fixed-effects treatment coefficients for secondary outcomes

outcome	estimate	std_error	p_value
RPE	0.5789	0.4237	0.1803
Motivation	0.4737	0.2905	0.1117
Energy	0.2895	0.3259	0.3803

0.40 HTE setup

We estimate treatment interactions on the primary TTE outcome. Because subgroup results are exploratory, we report Bonferroni-adjusted p-values across the three planned subgroup tests.

0.41 HTE models

Table 27: Exploratory HTE interaction results with Bonferroni correction

subgroup	interaction_term	estimate	std_error	p_value	p_value_bonferroni
Age	high_tempo:age_centered	0.2723	0.5090	0.5960	1
Gender (male vs not male)	high_tempo:male_indicator	5.5517	13.5212	0.6839	1
Activity level / fitness proxy	high_tempo:activity_level_or_0	0.8812	5.3176	0.8693	1

0.42 Design-Based Inference Check

The report closes the analytic sequence with randomization inference, which checks whether the observed paired effect is unusual under the sharp null implied by the randomization mechanism.

0.43 Observed statistic

Table 28: Observed statistic for randomization inference

quantity	value
Participants	38.0000
Observed paired mean difference (high - low)	14.3694

0.44 Randomization distribution

Table 29: Randomization inference summary

quantity	value
Simulations	10000.0000
Observed paired mean difference	14.3694
Two-sided RI p-value	0.0031
One-sided RI p-value (high > low)	0.0014

0.45 Figure

